COMP34812: Authorship Verification

Abdullah Sweesi, Hin Yui Jacob Yip
Department of Computer Science, University of Manchester

MANCHESTER 1824
The University of Manchester

Abstract

- **Authorship Verification (AV)** determines whether two sequences share a common author
- We propose Siamese Bi-LSTM and Cross-Encoder BERT models that integrate **semantic** context with a shared suite of 203-dimensional **stylometric** features to compare author 'fingerprints'
- Both transformer and non-transformer models significantly **outperformed** baseline methods

Introduction

- Authorship Verification (AV) has applications in forensic linguistics, plagiarism detection, and historical attribution [1, 2, 3]
- However, this field faces a critical challenge in cross-genre verification, in which one author may write across multiple themes
- Research has shown that the performance of Siamese architecture is sensitive to the method of semantic fusion [4]
- Cross-Encoder improves from Bi-Encoder by computing attention information between sub-tokens of two texts simultaneously instead of separately [5]

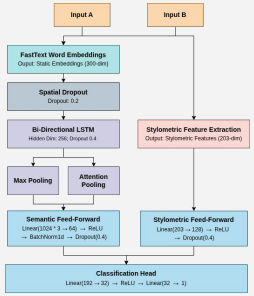

Figure 1: Siamese Bi-LSTM Architecture

Methodology

Siamese Bi-LSTM Model

- **Stylometry**: 203-dimensional vector (e.g. function words, POS trigrams, OOV, TTR) using absolute pairwise difference
- **Semantics**: FasText embeddings processed by a 3-layer Siamese Bi-LSTM with spatial dropout and dual (max/attention) pooling
- Tested various fusions to maximise authorial signal
- Late fusion of semantic and stylometric features in MLPs
- Hyperparameters fine-tuned over randomized search

Cross-Encoder BERT Model

- Applies the same suite of 203 stylometric features
- **Semantics**: Contextual embeddings processed by a BERT-base-cased transformer followed by a feed-forward neural network to extract shared features for heads
- **Global Author Head**: Compare all extracted features simultaneously obtained from joint encoding
- **Neural Style Head**: Compare high-level semantic patterns
- Weighted fusion of similarity scores from heads of different perspectives

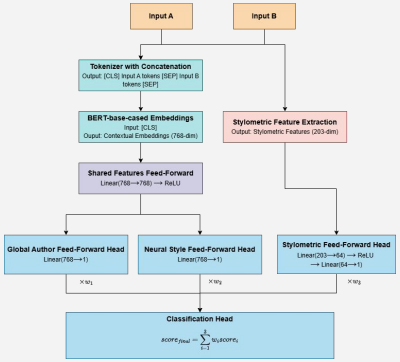

Figure 2: Cross-Encoder Architecture

Results

Solution	Accuracy	Macro Precision	Macro Recall	Macro F1 Score	MCC
Baseline LSTM	0.623	0.623	0.623	0.623	0.245
Siamese Bi-LSTM	0.738	0.738	0.738	0.738	0.476
Baseline BERT	0.785	0.785	0.786	0.785	0.571
CE BERT	0.832	0.833	0.833	0.832	0.666

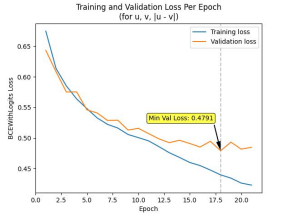

Figure 3: Training and Validation Loss in Siamese Bi-LSTM

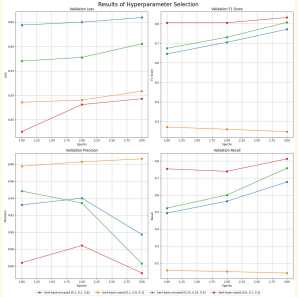

Figure 4: Validation Metrics in Cross-Encoder BERT

Conclusion

- Both architectures significantly **outperformed** baseline models across all metrics
- Siamese Bi-LSTM achieved a 0.115 gain in Macro-F1 over baseline LSTM
- Experimental results indicate that the optimal method of semantic fusion is $[\mathbf{u}, \mathbf{v}\ |\mathbf{u} - \mathbf{v}|]$, corroborating SBERT findings [4]
- Spatial dropout and pooling strategies greatly enhanced generalisation and reduced overfitting
- Cross-Encoder BERT achieved a 0.047 gain in Macro-F1 over baseline BERT
- Experimental results indicate that combining texts before embedding gives performance improvement, corroborating Cross-Encoder findings [5]
- Feed-forward neural network extracts shared features, greatly enhanced generalisation and distilled a unified representation for following, specialised heads
- Future work: feature significance for interpretability, exploring triplet/contrastive loss, testing against adversarial stylometry, diverse classification heads

References

- [1] Juola, P., 2021. Verifying authorship for forensic purposes: A computational protocol and its validation. Forensic Science International, 325, p.110824. https://doi.org/10.1016/j.forsciint.2021.110824
- [2] Enriquez, D., et al., 2022, May. Authorship verification for hired plagiarism detection. In Proceedings of the 9th international conference on applied computing & information technology (pp. 19-24). https://doi.org/10.1145/3543895.3543928
- [3] Stover, J.A., et al., 2016. Computational authorship verification method attributes a new work to a major 2nd century african author. Journal of the Association for Information Science and Technology, 67(1), pp.239-242. https://doi.org/10.1002/asi.23460
- [4] Reimers, N. and Gurevych, I., 2019, November. Sentence-bert: Sentence embeddings using siamese bert-networks. In Proceedings of the 2019 conference on empirical methods in natural language processing and the 9th international joint conference on natural language processing (EMNLP-IJCNLP) (pp. 3982-3992). https://doi.org/10.48550/arXiv.1908.10084
- [5] Konstantinou, S., Li, J. and Zinonos, A., 2022. Different Encoding Approaches for Authorship Verification. In CLEF (Working Notes) (pp. 2532-2540). https://ceur-ws.org/Vol-3180/paper-206.pdf